Do Now · retrieval

LMC branching and debugging

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What does STA do?

Q2

What does LMC HLT do?

Q3

Why must a loop's counter change?

Q4

Full adder: inputs A=1, B=0, carry-in=1. Find sum and carry-out.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Understand — BRP tests whether ACC is non-negative; BRZ tests zero; BRA is unconditional.

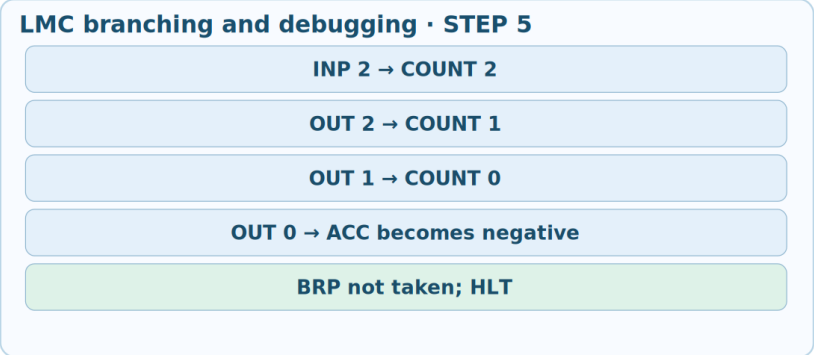
Core idea 2

Trace — Load the counter, output it and subtract one. Test BRP immediately: save the new value and loop only when it is non-negative.

Core idea 3

Explain — After outputting 0, SUB ONE underflows; BRP is not taken, so HLT runs without storing a negative mailbox value.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

BRA — Branch Always

BRZ — Branch if Zero

BRP — Branch if non-negative

mailbox — LMC memory cell

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

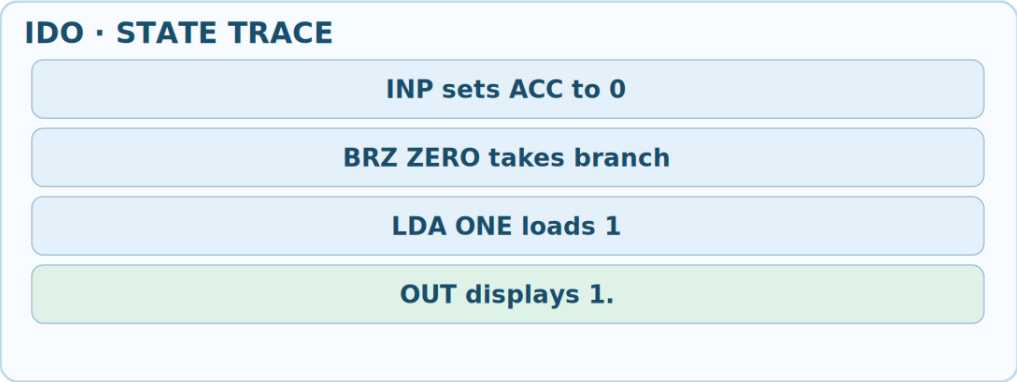
Problem

Trace BRZ when input is 0.

Follow the worked method

1. Predict first — INP sets ACC to 0
2. Check next state — LDA ONE loads 1
3. Check next state — OUT displays 1.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict first
INP makes ACC 4

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Define BRA, BRZ and BRP and explain their effect on PC. [4]

Question 2

Input 3: trace INP; BRZ STOP; SUB ONE; OUT; HLT; ONE DAT 1; STOP HLT. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Explain why an unconditional backward branch without a tested exit can run forever, and provide a safe counter-based solution. [6]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

An unconditional backward branch always terminates.

Use key vocabulary from memory

1. Explain BRA in your own words.

2. Explain BRZ in your own words.

3. Explain BRP in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.